Solar ESS Co. Ltd.
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ROLLER 600 Roller Heat Press Electrical Supply & Motor Drive Recommendations

Rev01 · 10 November 2026 · 380 V 3-phase · 56 kW machine

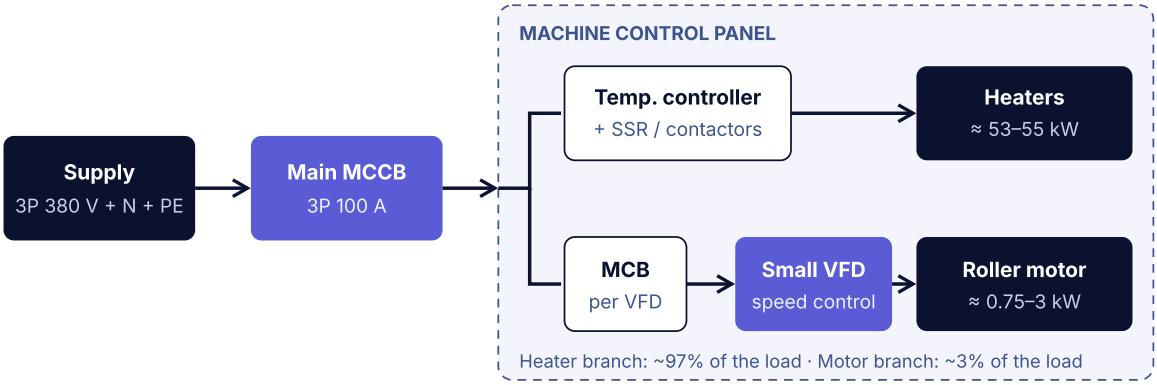
The 56 kW rating is **almost all heaters**. Only a small motor turns the roller. You need a **100 A supply** and, at most, a **small VFD** for the roller motor. A 55–75 kW soft starter or VFD is **not needed**.

HEATERS
≈ 53–55 kW
Heat the drum. Switched by the temperature controller via SSR/contactors.

ROLLER MOTOR
≈ 0.75–3 kW
Turns the drum. Needs variable speed to set dwell time.

TOTAL CURRENT
≈ 85 A
56 kW ÷ (√3 × 380 V), mostly resistive load.

1 How the power is distributed



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2 Recommended specification

Item	Specification	Note
Main breaker	**MCCB 3P 100 A**, Icu ≥ 25 kA	Use 125 A only if the maker's manual requires it.
Supply cable	**CV 4 × 35 mm² + G 16 mm²**	Copper XLPE. Size up for runs over ~50 m.
Heater control	Supplied with the machine	Temperature controller + SSR/contactors. No soft starter, no VFD.
Roller motor drive	Small VFD, one size above motor kW	Only if the machine has none. Heavy duty, sensorless vector.
VFD protection	MCB / fuse per VFD manual	e.g. 16–20 A for a 2.2 kW drive.
Earthing	Machine frame, panel and VFD to one earth bar	Required for safety and for the digital controls.

3 Do and don't

✓ Do

- Check the motor nameplate (kW, A, V) first.
- Check whether a speed inverter is already in the panel; most are.
- Use a VFD for roller speed; it sets the dwell time on the heat.
- Size the supply from total current (≈ 85 A).

× Don't

- Buy a 55–75 kW soft starter: heaters don't need one, and it can't control speed.
- Buy a 55–75 kW VFD: the motor is only a few kW.
- Put heaters on a VFD or soft starter.
- Run control cables in the same duct as power cables.

4 Suggested VFDs (only if the machine has none)

Model range	Tier	Why
Delta MS300	Value	Widely available in Thailand; good low-speed torque.
Schneider ATV320	Mid	Reliable; built-in Safe Torque Off.
ABB ACS380	Premium	Machine-control drive; easy commissioning.
INVT GD20	Budget	Low cost; adequate for a simple roller drive.

Example: 1.5 kW motor → 2.2 kW VFD (heavy-duty rating). Final model to be confirmed from the motor nameplate.

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5 Step-by-step plan

1. **Ask the machine seller:** motor kW/A; built-in inverter (yes/no); heating method (electric or oil drum); neutral needed; recommended main breaker; wiring diagram.
2. **Prepare the supply:** MCCB 3P 100 A and CV 4 × 35 mm² + G 16 mm² to the machine. This can start now.
3. **Choose the drive:** only if no inverter is fitted, select a VFD one size above the motor kW.
4. **Install:** separate power and control ducts; bond everything to one earth bar; label the panel.
5. **Commission:** enter motor nameplate data in the VFD, set ramp times and speed range, test the temperature cut-off.

Figures are engineering estimates from the published machine rating (380 V, 3-phase, 56 kW). Confirm against the manufacturer's data before purchase.